



School of Social Sciences and Philosophy
Assignment Submission Form

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Signed: Monjish Sarkar

Date: 13/04/26

1.Introduction

The objective of this paper is to examine the impact of oil price volatility on macroeconomic and financial aggregates in the United States. Oil is a key input in production, transport and global trade, so as a result instability in oil markets influence the economy through numerous channels. Rise in oil market volatility may increase uncertainty for firms and households, discourage investment and spending, and affect financial market sentiment. Therefore, despite the difficulty in pinning down the direct effect of oil price changes, oil volatility is potentially important. In other words, what is of crucial importance is not whether oil prices rise or fall but whether oil markets become unstable and uncertain.

The United States is the country under consideration in this paper. This is because the US provides an appropriate setting for this question since oil-market volatility can plausibly influence both the real economy and financial markets. On the macroeconomic side, an increase in oil market volatility may weaken production and activity by raising input uncertainty and encouraging firms to postpone decisions. On the financial side, stock market returns respond immediately if investors interpret oil-market inconsistencies as a sign of higher risk or weaker future economic performance. Since the topic addresses both macroeconomic and financial aggregates, the empirical analysis is designed to capture one variable from each side of the economy.

The question addressed in this paper is: “how does an oil volatility shock affect US real activity and stock market returns over time?” The focus here is on dynamic transmission rather than a simple contemporaneous correlation. In a time-series context, the effect of a shock often unfolds over several periods, and an appropriate empirical framework should therefore accommodate the variables to depend on their past values and the past values of other variables in the system.

The natural starting point of this topic therefore is a multivariate time-series model. Initially, I explored richer model systems including inflation and, in some specifications, the federal funds rate. However, once lag selection, stability checks, and residual serial correlation diagnostics were conducted and compared across specifications, the broader systems behaved poorly. The main issue was the presence of residual serial autocorrelation. Though their poor performance did not necessarily mean that the models were impossible to estimate, it did mean that they were not the cleanest and most defensible compact specification for this paper. I then tried to determine whether the sample itself was contributing to the problem. The early half of the sample includes the Global Financial Crisis and other extreme disturbances, which was probably making it difficult for one linear monthly system to capture the data cleanly.

As a result, the final specification that I chose to use was a more parsimonious post-2010 three-variable VAR(2) including oil volatility, growth in industrial production and S&P 500 returns. To strengthen the analysis, I also incorporate a recursive structural VAR on the same system to check if imposing a simple structural identification changes the qualitative response patterns.

The rest of the paper is organised as follows. Section 2 will describe the data and variable construction; Section 3 outlines the empirical approach and explains the final model choice; Section 4 presents the empirical results and interprets the dynamic responses while Section 5 mentions limitations and policy implications.

2.Data Description

The analysis uses monthly US data. The raw series were collected from publicly available source FRED and converted into a common monthly frequency suitable for time-series estimation. The full cleaned data set begins in 2007, but the preferred baseline specification uses a post-2010 sample. This restriction was motivated by econometric considerations. The period before 2010 which contains the Global Financial Crisis introduces exceptionally large disturbances which makes it difficult for a compact linear VAR to fit the data cleanly. By focusing on the post crisis-period, the model is estimated over a more stable environment while still covering an economically relevant and important period that includes both calm and volatile-oil market episodes.

There are three variables that are used in the final baseline specification. The first one is oil price volatility, denoted by ovx_level_t . This is measured by using the OVX crude-oil volatility index. It is used to capture the oil-market uncertainty rather than the oil price level itself. It also serves as the main shock variable in the model. Conceptually, this is the variable whose impact on the rest of the system; this paper is interested in tracing.

The second variable is industrial production growth, denoted by ip_growth_t . Industrial production is used as the macroeconomic aggregate because it is available at monthly frequency and provides a timely measure of real activity. Instead of using the raw industrial production index, I transformed it into log growth rate in percent;

$$ip_growth_t = 100[\ln(INDPRO_t) - \ln(INDPRO_{{t-1}})].$$

This transformation makes the series more appropriate for time series analysis and improves interpretability.

S&P 500 returns is the third variable and is denoted by $sp500_rett$. This serves as the financial aggregate variable. Daily S&P 500 data are first converted to monthly frequency by keeping the last trading day of each month. Monthly log returns are then constructed as:

$$sp500_ret_t = 100[\ln(SP500_t) - \ln(SP500_{{t-1}})].$$

Again, using returns rather than the raw stock index level is more natural in a time-series setting and gives a clearer measure of short-run financial-market performance.

3. Empirical Approach

The empirical analysis is majorly based on Vector Autoregression (VAR), which is suitable because the question of interest is dynamic, involves multiple time series and may influence one another over time. In this specific system, each variable is explained by its own lagged values and the lagged values of the other variables. The VAR therefore makes sense for tracing how an oil-volatility shock is transmitted to real activity and financial-market returns over subsequent months.

Let, $X_t = [\text{ovx_level}_t, \text{ip_growth}_t, \text{sp500_ret}_t]'$

Then, the reduced-form VAR of order p is written as

$$X_t = c + A_1 X_{t-1} + A_2 X_{t-2} + \dots + A_p X_{t-p} + u_t,$$

Where c is a vector of constants, $A_1, \dots, A_p$ are coefficient matrices, and u_t is a vector of reduced form innovations.

A number of alternative models were compared before the final specification was selected. Broader systems included inflation and, in some specifications, the federal funds rate was considered first as they seemed natural extensions of the topic. However, even though they often satisfied the stability condition, the larger VARs were persistently characterized by residual serial autocorrelation across plausible lag choices. I then tried to investigate whether the sample itself was causing the problem. Restricting the sample to post 2010 period helped to substantially improve model performance, especially by excluding the most turbulent years of the Global Financial Crisis from the estimation window.

Lag length was chosen by combining standard lag order criteria with diagnostic checks. In the final three-variable system, candidate models with lags 2 and 4 were considered. Both were found to be stable but VAR(2) proved to lead to the cleaner overall specification in terms of residual diagnostics.

After estimating the baseline reduced form VAR, I analysed the effects of oil volatility shock using Orthogonalized Impulse Response Functions (IRF's). Since orthogonalized IRFs depend on the ordering of variables, the baseline ordering is

$\text{ovx_level}_t \rightarrow \text{ip_growth}_t \rightarrow \text{sp500_ret}_t$.

This ordering implies that oil volatility is contemporaneously prior within the month, industrial production growth can respond contemporaneously to oil volatility and stock returns can respond contemporaneously to both. I also used reverse ordering to assess the sensitivity of the results to this assumption. Additionally, I computed the forecast error variance decompositions (FEVDs) in

order to understand how much of the forecast error variance of industrial production growth and stock returns can be linked to oil-volatility shocks over different horizons.

Finally, I estimate a recursive structural VAR on the same system as a short extension. I do this to check whether imposing a simple recursive structural interpretation materially changes the qualitative response patterns obtained from the baseline VAR.

4. Empirical Results and Interpretation

4.1 Model Selection and Diagnostic Comparison -

Specification	Sample	Stable?	LM p-value lag 1	LM p-value lag 2	Comment
4-variable VAR(2): (ovx_level, ip_growth, infl_mom, sp500_ret)	Full sample	Yes	0.36837	0.00033	Fails at lag 2
4-variable VAR(4): (ovx_level, ip_growth, infl_mom, sp500_ret)	Full sample	Yes	0.01887	0.13449	Fails at lag 1
4-variable VAR(2): (ovx_level, ip_growth, infl_mom, sp500_ret)	Post-2010	Yes	0.03195	0.00005	Fails at both lags overall
4-variable VAR(4): (ovx_level, ip_growth, infl_mom, sp500_ret)	Post-2010	Yes	0.02236	0.02141	Fails at both lags
3-variable VAR(2): (ovx_level, ip_growth, sp500_ret)	Post-2010	Yes	0.94691	0.79309	Preferred specification

3-variable VAR(4): (ovx_level, ip_growth, sp500_ret)	Post-2010	Yes	0.01122	0.01534	Worse than VAR(2)
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Table 1: Diagnostic comparison across different candidate VAR specifications

The specification of choice becomes straightforward from Table 1. Restricting the sample to post-2010 period improves performance, but this alone does not solve the issue of residual serial autocorrelation in the 4-variable system. Once inflation is removed and attention is focused on oil volatility, industrial production growth and stock returns, the VAR behaves much more cleanly. The preferred model is therefore selected on econometric grounds rather than because it is simply smaller.

4.2 Baseline Impulse Response Functions -

The dynamic implications of the preferred specification are examined using orthogonalized impulse response functions.

Figure 1. Orthogonalized impulse responses to an oil-volatility shock (baseline ordering)

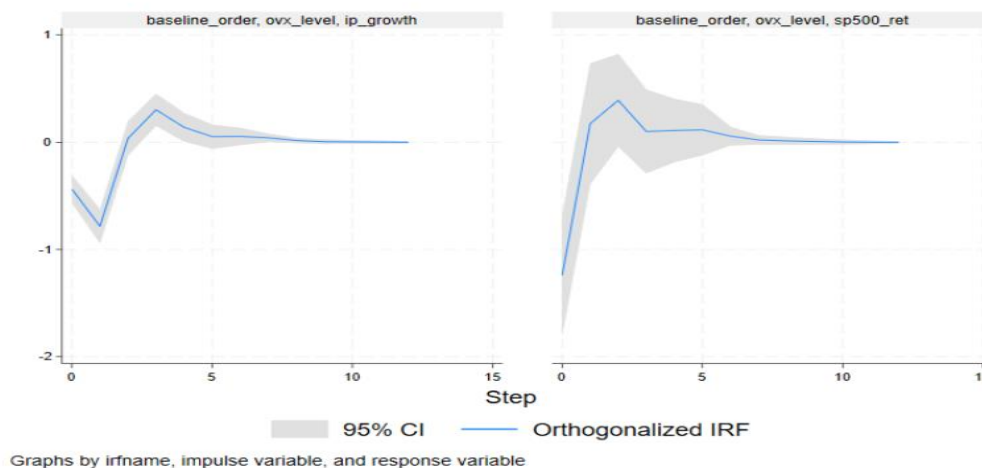
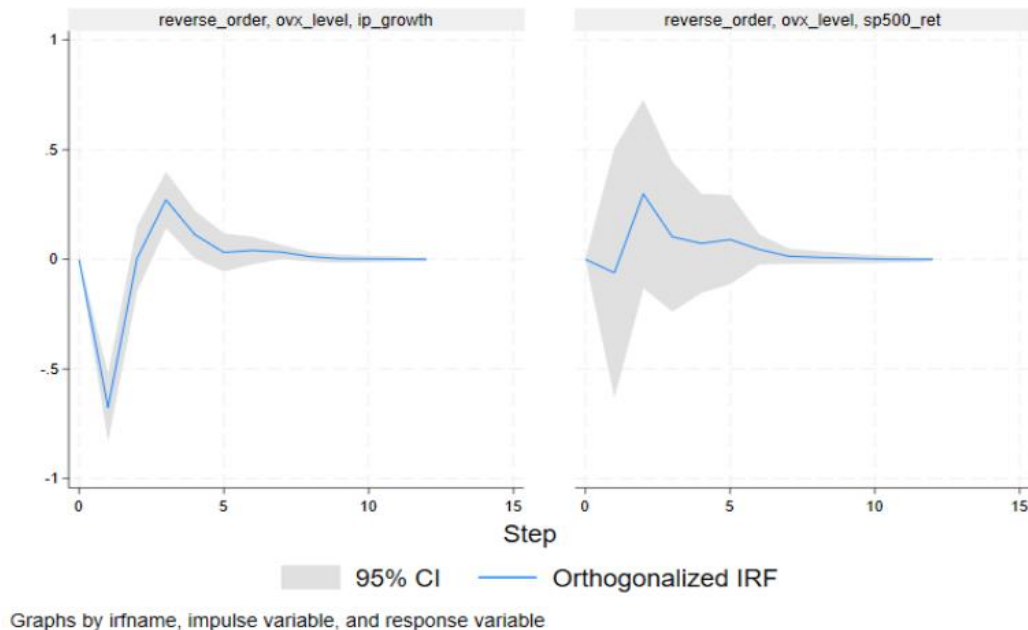


Figure 1 reports the orthogonalized impulse responses from the VAR(2). Following a shock to oil volatility, industrial production growth falls immediately and remains negative in the consequent report before recovering, turning slightly positive and gradually converging back towards zero. From this I infer that higher oil market uncertainty weakens real activity in the short run, but that effect is temporary rather than permanent. Stock returns on the other hand display a much sharper impact response. They fall strongly on impact and then recover over subsequent periods. The financial market response is therefore much more pronounced than industrial production growth.

This is economically plausible as well, since asset prices generally tend to react quickly to uncertainty shocks than real activity does.

4.3 Robustness to Alternative Ordering

Figure 2. Orthogonalized impulse responses under reverse ordering



Since orthogonalized impulse responses depend on variable ordering, I also estimated the model under a reverse ordering as a robustness check. The broad pattern remains intact. Industrial production growth still shows an initial decline followed by recovery, while stock returns continue to display a short run negative response and subsequent normalization. Even though the magnitudes do change somewhat, especially for stock returns, qualitative conclusions do not. This means that the main findings are not solely driven by one specific Cholesky ordering.

4.4 Forecast Error Variance Decomposition -

Forecast error variance decomposition is then used to assess the relative quantitative importance of oil-volatility shocks over different horizons.

Figure 3. Forecast error variance decomposition (baseline ordering)

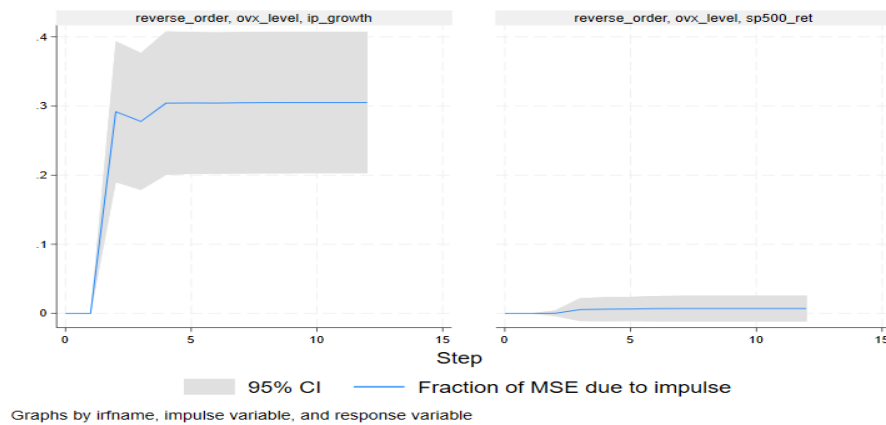
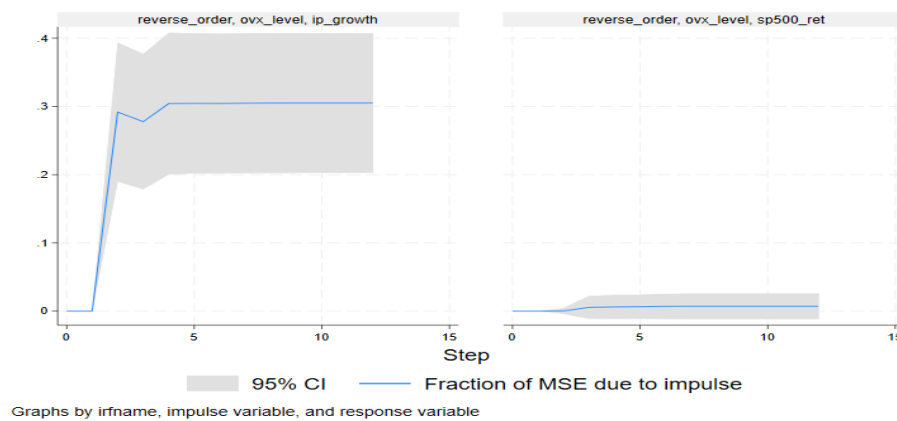


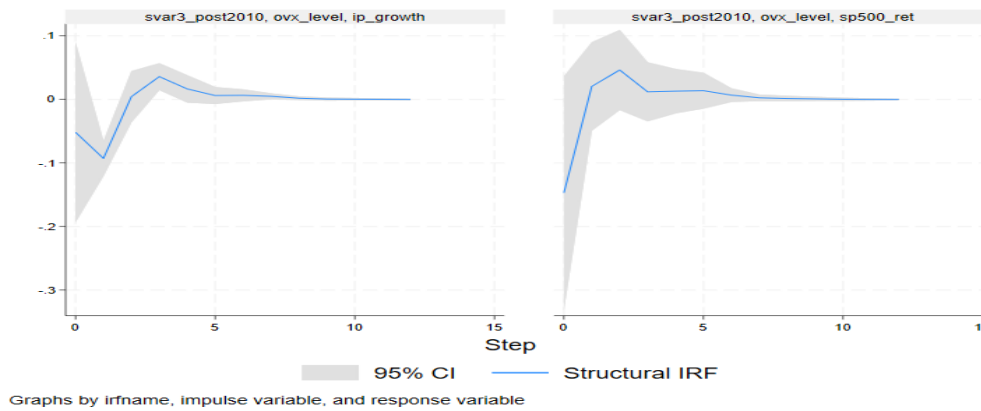
Figure 4. Forecast error variance decomposition (reverse ordering)



The forecast error variance decompositions provide additional insight into the relative importance of oil volatility shocks. In the baseline ordering, oil-volatility shocks explain a meaningful share of forecast error variance in industrial production growth over time, where the contribution to stock return variance is smaller. Under reverse ordering, the broad conclusion remains similar for industrial production growth, while contribution to stock returns becomes even smaller. Therefore, the evidence from FEVD supports the view that oil volatility shocks are quantitatively more important for explaining variation in industrial production growth than for explaining variation in stock returns over longer horizons, even though the immediate stock-market response is stronger.

4.5 Recursive SVAR extension -

Figure 5. Structural impulse responses from the recursive SVAR



Finally, I estimated a recursive SVAR on the same post-2010 3-variable system. The structural impulse responses are qualitatively similar to those from the reduced-form VAR. Industrial production growth again declines in the short run and then recovers, while stock returns show a sharp negative impact response followed by normalization. I therefore treat the SVAR as supporting evidence rather than as the main model. Taken together, the baseline IRFs, reverse-ordering check, FEVD results and SVAR extension all point in the same direction: oil-volatility shocks are associated with weaker short-run real activity and weaker short-run stock-market performance in the United States.

5.Limitations and Policy Implications -

A few limitations must be acknowledged. First, broader specifications including inflation and the federal funds rate performed less well diagnostically and were therefore not included in the main model. The preferred specification is parsimonious by design. Second, the analysis deliberately focuses on the post-2010 period to avoid the most unstable period associated with the Global Financial Crisis. Although this improves diagnostics, broader generalization is limited. Third, recursive ordering helps identify shocks, but responses for stock returns remain somewhat ordering sensitive.

The results carry meaningful economic intuition and relevant policy implications. Since oil market volatility shocks tend to weaken real activity and generate an immediate response in the financial markets, policy makers should not view oil market instability as only a narrow commodity market problem. Instead, it may act as an early indicator for macro-financial stress. In the context of monetary and macroeconomic authorities, this implies that periods of heightened oil market volatility may warrant closer monitoring of output conditions and financial market sentiment, especially when uncertainty is already elevated. Stronger and faster responses of stock returns also

mean that financial markets transmit oil market related uncertainty more rapidly than the real economy which is important for interpreting short-run market movements and for communicating policy in volatile periods. More broadly, the results suggest that policy frameworks concerned with economic stabilization should pay attention not only to oil-price levels, but also to volatility in oil markets as a separate source of short-run disturbance.

References

- Enders, W. (2014). Applied Econometric Time Series. 4th edn. Hoboken, NJ: Wiley.
- Federal Reserve Bank of St. Louis (n.d.) CBOE Crude Oil ETF Volatility Index [OVXCLS]. FRED, Federal Reserve Bank of St. Louis. Available at: <https://fred.stlouisfed.org/series/OVXCLS> (Accessed: 13 April 2026).
- Federal Reserve Bank of St. Louis (n.d.) Industrial Production: Total Index [INDPRO]. FRED, Federal Reserve Bank of St. Louis. Available at: <https://fred.stlouisfed.org/series/INDPRO> (Accessed: 13 March 2026).
- Investing.com (n.d.) S&P 500 Historical Data. Available at: <https://www.investing.com/indices/us-spx-500-historical-data> (Accessed: 13 March 2026).
- Kilian, L. and Park, C. (2009) 'The impact of oil price shocks on the U.S. stock market', International Economic Review, 50(4), pp. 1267-1287.